

Lab Work 4
Course Title: Information Security Laboratory/Project
Course Code: 0714 02 CSE 4108
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Implementation of Diffie-Hellman Key Exchange

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1. print("      DIFFIE-HELLMAN KEY EXCHANGE")
2. q = int(input("Enter the prime number q: "))
3. while True:
4.     if q < 2:
5.         prime = False
6.     else:
7.         prime = True
8.         i = 2
9.         while i*i <= q:
10.            if q % i == 0:
11.                prime = False
12.                break
13.            i += 1
14.     if prime:
15.         break
16.     q = int(input(str(q) + " is not prime, enter q again: "))
17.
18. alpha = int(input("Enter the primitive root alpha: "))
19. while True:
20.     if alpha < 2 or alpha >= q:
21.         primitive = False
22.     else:
23.         values = []
24.         value = 1
25.         primitive = True
26.         i = 1
27.         while i < q:
28.             value = (value*alpha) % q
29.             if value in values:
30.                 primitive = False
31.                 break
32.             values.append(value)
33.             i += 1
34.     if primitive:
35.         break
36.     alpha = int(input(str(alpha) + " is not a primitive root of " + str(q)
+ ", enter alpha again: "))
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37.
38.xa = int(input("Enter the private key of A (XA): "))
39.while xa < 1 or xa >= q:
40.    xa = int(input("XA must be between 1 and q-1, enter XA again: "))
41.
42.xb = int(input("Enter the private key of B (XB): "))
43.while xb < 1 or xb >= q:
44.    xb = int(input("XB must be between 1 and q-1, enter XB again: "))
45.
46.ya = 1
47.i = 0
48.while i < xa:
49.    ya = (ya*alpha) % q
50.    i += 1
51.
52.yb = 1
53.i = 0
54.while i < xb:
55.    yb = (yb*alpha) % q
56.    i += 1
57.
58.ka = 1
59.i = 0
60.while i < xa:
61.    ka = (ka*yb) % q
62.    i += 1
63.
64.kb = 1
65.i = 0
66.while i < xb:
67.    kb = (kb*ya) % q
68.    i += 1
69.
70.print("Global Public Elements")
71.print("q =", q)
72.print("alpha =", alpha)
73.print("User A Key Generation")
74.print("Private XA =", xa)
75.print("Public YA =", ya)
76.print("User B Key Generation")
77.print("Private XB =", xb)
78.print("Public YB =", yb)
79.print("Exchange")
80.print("A sends YA =", ya, "to B")
81.print("B sends YB =", yb, "to A")

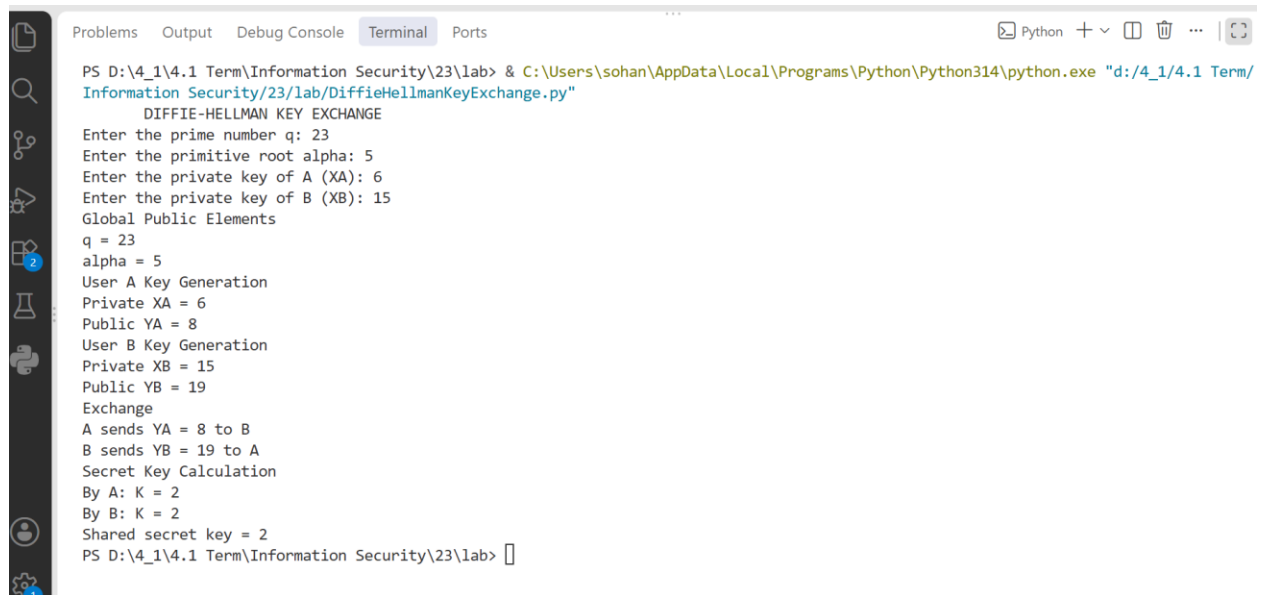
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82.print("Secret Key Calculation")
83.print("By A: K =", ka)
84.print("By B: K =", kb)
85.if ka == kb:
86.    print("Shared secret key =", ka)
87.else:
88.    print("Key exchange failed")

```

Output:



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PS D:\4_1\4.1 Term\Information Security\23\lab> & C:\Users\sohan\AppData\Local\Programs\Python\Python314\python.exe "d:/4_1/4.1 Term/Information Security/23/lab/DiffieHellmanKeyExchange.py"
DIFFIE-HELLMAN KEY EXCHANGE
Enter the prime number q: 23
Enter the primitive root alpha: 5
Enter the private key of A (XA): 6
Enter the private key of B (XB): 15
Global Public Elements
q = 23
alpha = 5
User A Key Generation
Private XA = 6
Public YA = 8
User B Key Generation
Private XB = 15
Public YB = 19
Exchange
A sends YA = 8 to B
B sends YB = 19 to A
Secret Key Calculation
By A: K = 2
By B: K = 2
Shared secret key = 2
PS D:\4_1\4.1 Term\Information Security\23\lab>

```

Figure: Diffie-Hellman Key Exchange Output